

Modeling: MultiModal AI — Homework 4

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In this homework, you will explore **Reinforcement Learning for Vision-Language Models**. Specifically, you will implement and train a VLM using **Group Relative Policy Optimization (GRPO)**, a recent RL algorithm for aligning language models with reward signals.

Part 1: Reading & Reflection (20 points)

Required Reading

1. [DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models](#) (Sections 1, 3, and 4)
 2. [The Illustrated GRPO](#) (Full post)
-

Questions

1. **GRPO vs PPO:** GRPO removes the need for a learned value function (critic). Explain how GRPO estimates the baseline for advantage computation without a critic, and what tradeoffs this introduces.

Instead of using a learned value function to calculate the Advantage of the response, GRPO uses the average reward across a batch of sampled responses from the initial policy. The average is expected to capture both bad and good responses, and thus, if a response is better than average, the RL update will encourage the updated policy towards that response. The advantages of this new approach is focused around the benefits of not training a separate learned value function. First, that requires data, and can be susceptible to overfitting. Open-ended responses can be nuanced, so learning the value function can be difficult. Also, the learned value function training might use more compute/memory. However, the GRPO approach requires more compute in sampling outputs and calculating rewards. The main problem comes from the nature of the responses: an initial policy that is biased towards a specific type of output might take longer to converge on difficult tasks because the baselines are not as informative as the learned value function can be. In other words, if all responses are fine, but not great, or consistently bad, there might be slow progress for that task.

2. **Reward Design:** Discuss the difference between using a learned reward model vs. a rule-based reward function (e.g., checking if an answer matches ground truth). What are the risks of reward hacking in each case?

A rule-based reward function is deterministic, which makes it easier to debug and understand, and there is little noise. However, there are many cases where open-ended responses like those outputted by LLMs might contain nuance that evade the rule-based reward function. For example, we might ask an LLM to give us the color of a firetruck, with the expected correct

answer to be red. If the LLM instead responds with crimson, a rule-based reward function might not have the expressiveness to understand that this is still a good response, and therefore the reward would be low. On the other-hand, a learned reward model might be able to understand the similarity between crimson and red, and therefore appropriately score the response. Similar issues arise in formatting, where diverging formats might cause rule-based functions to misscore outputs. However, when training a reward model, while we add nuance and expressivity to the reward function, we need lots of high-quality data (usually human feedback data), which is expensive to obtain and train on. Also, we might hallucinate rewards, or the model might become susceptible to distribution shift between reward model training and RL training..

3. **SFT vs. GRPO:** In Homework 3, you fine-tuned a VLM using supervised fine-tuning (SFT) with LoRA. Compare the SFT approach with GRPO training: how does the training signal differ, and when would each be preferred?

In the SFT approach, we assume that the answers to prompts are clear and verifiable, and that we have access to data that captures those correct answers. This training signal is essentially that of mimicking the SFT dataset. However, we might prefer GRPO when we have more nuance in outputs, or we require reasoning, as there will most likely be no real ground-truth dataset for those types of responses. In other words, there are some tasks where there are no "right" and "wrong" answers, instead, levels of preferences over all possible answers that the model must learn. This also allows the models to explore new policies, potentially improving on pretraining or SFT policies by encouraging more preferable responses.

Part 2: Implementing and Training GRPO (100 points)

Problem 1: GPU Verification and Library Installation

```
In [1]: !pip install transformers accelerate bitsandbytes pillow torch torchvision trl peft data

import torch

print("PyTorch version:", torch.__version__)
print("CUDA available:", torch.cuda.is_available())
print("CUDA device count:", torch.cuda.device_count())
if torch.cuda.is_available():
    print("GPU name:", torch.cuda.get_device_name(0))

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
t = torch.randn(2, 3, device=device)
KEY = 73
cipher_bytes = [14, 27, 25, 6, 105, 32, 58, 105, 8, 37, 37, 105, 16, 38, 60, 105, 7, 44,

if t.is_cuda:
    cipher = torch.tensor(cipher_bytes, dtype=torch.uint8, device=device)
    decoded = cipher ^ KEY
    secret_word = "".join(chr(c) for c in decoded.cpu().tolist())
    print(f"\nGPU check passed! Secret word: {secret_word}")
else:
    print("\nNo GPU detected. Please switch to an A100 runtime.")
```

	60.7/60.7 MB	38.8 MB/s	eta 0:00:00
	697.4/697.4 kB	55.9 MB/s	eta 0:00:00
	527.0/527.0 kB	54.5 MB/s	eta 0:00:00
	47.6/47.6 MB	53.6 MB/s	eta 0:00:00
	36.3/36.3 MB	57.9 MB/s	eta 0:00:00

PyTorch version: 2.10.0+cu128
 CUDA available: True
 CUDA device count: 1
 GPU name: NVIDIA A100-SXM4-40GB

GPU check passed! Secret word: GRP0 is All You Need

Problem 2: Prepare Your Dataset (10 points)

You will **reuse the dataset you prepared in Homework 3** (or Homework 1/2). The data format remains the same:

```
mmai-data/
├── images/
│   ├── image_01.jpg
│   └── ...
└── data.jsonl
```

Each line in `data.jsonl` should be a JSON object with:

```
{
  "image": "images/1.jpg",
  "question": "What animal is in this image?",
  "answer": "cat"
}
```

If you do not have your own dataset, the default Google Drive link below will download a small example dataset so you can complete the homework. You will lose some points for not using your own data, but you can still finish every problem.

As in HW3, you should have a **train/test split**. The test images should not be used during training.

In HW3, you used supervised fine-tuning (SFT), which directly trains the model to produce the correct answer. In this homework, we use **reinforcement learning**: the model generates its own answers, receives a reward signal, and updates its policy to maximize future reward.

To enable the RL training loop, we append an instruction after each user question asking the model to **think step-by-step** and then place its final answer after `Answer:` . This structured output lets us automatically extract and evaluate the model's prediction.

READ HERE: ROHAN'S DATASET

I do not have a good dataset for this homework because the answers to the questions in my chest x-ray dataset that I used for HW1-3 are very open-ended, so measures like "accuracy" are going to almost always be 0. Instead, I built out a new dataset based on movie posters. The targets for the questions are genres and decades for different posters. I scraped this data from an IMDB

clone called TMDb, and it contains 300 images from 6 genres, 50 each. I also split this up into a train-test split so that we have held-out images for testing later. I hope I can get full points.

```
In [2]: import os, shutil, zipfile, json
        from pathlib import Path

        # =====
        # ##### CHANGE ME #####
        # =====
        # Upload your dataset as a zip file to Google Drive, then
        # replace the URL below with your own Google Drive share link.
        #
        # If you do not have your own dataset, leave the default URL.
        # The default dataset is a small example so you can still
        # complete the homework.

        GDRIVE_SHARE_URL: str = "https://drive.google.com/file/d/1P88-wAHg137qGFggbi2I2_3V6xogBn

        # =====
        # ##### END CHANGE ME #####
        # =====

        DATA_ROOT = Path("/content/movie-dataset-creation")
        ZIP_PATH = Path("/content/mmai-data.zip")

        if not DATA_ROOT.exists():
            if GDRIVE_SHARE_URL:
                import gdown
                gdown.download(GDRIVE_SHARE_URL, str(ZIP_PATH), quiet=False, fuzzy=True)
                with zipfile.ZipFile(ZIP_PATH) as zf:
                    zf.extractall(".")
                ZIP_PATH.unlink(missing_ok=True)
            else:
                print("Please set GDRIVE_SHARE_URL above, or manually upload mmai-data/ folder.")

        # Verify dataset
        if DATA_ROOT.exists():
            jsonl_path = DATA_ROOT / "data.jsonl"
            with open(jsonl_path) as f:
                lines = [json.loads(l) for l in f if l.strip()]
            print(f"Dataset loaded: {len(lines)} samples")
            print(f"Sample: {lines[0]}")
        else:
            print("Dataset not found. Please upload your data.")
```

```
Downloading...
From: https://drive.google.com/uc?id=1P88-wAHg137qGFggbi2I2_3V6xogBn_i
To: /content/mmai-data.zip
100%|██████████| 12.0M/12.0M [00:00<00:00, 157MB/s]
Dataset loaded: 1797 samples
Sample: {'image': 'image/0000.jpg', 'question': 'What genre does this movie belong to?',
'answer': 'action'}
```

Problem 3: Understanding GRPO (15 points)

Before implementing GRPO, let us walk through the key ideas.

Background: From PPO to GRPO

Standard RLHF methods like **PPO (Proximal Policy Optimization)** require four models: a policy, a reference model, a reward model, and a value model (critic). GRPO simplifies this by **removing the value model entirely**, using the group of completions generated for the same prompt to estimate a baseline.

The GRPO Algorithm

Given a prompt q , we sample G completions $\{o_1, o_2, \dots, o_G\}$ from the current policy π_θ . Each completion receives a reward r_i .

Step 1: Compute the Group Advantage

$$\hat{A}_i = \frac{r_i - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r}) + \epsilon}$$

where $\mathbf{r} = [r_1, \dots, r_G]$ are the rewards for all completions of the same prompt.

Step 2: Clipped Surrogate Loss

$$L_{\text{GRPO}}(\theta) = -\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(\rho_{i,t} \hat{A}_i, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right)$$

where $\rho_{i,t} = \frac{\pi_\theta(o_{i,t}|q, o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q, o_{i,<t})}$ is the importance sampling ratio.

Step 3 (Optional): KL Regularization

$$L_{\text{total}} = L_{\text{GRPO}} + \beta \cdot \mathbb{D}_{\text{KL}}[\pi_\theta || \pi_{\text{ref}}]$$

Questions to Answer:

1. In your own words, explain why the group-based advantage normalization works as a baseline. What happens when all G completions for a prompt receive the same reward?

In general, the idea of advantage is to calculate the preference of one response over all others. By taking the group-based mean as the baseline, we are saying that a response is better or worse than the average response across a large group of options, which is the same intuition. Thus, we are able to forgo an explicit advantage model (critic), instead using the difference between the output's reward and the average response's reward. When we train via backprop, this increases probabilities of responses better than the baseline and vice versa. When all completions receive the same reward, there is no adjustment to the policy model, as the advantage of all options is 0.0, and thus there is no gradient update.

2. What is the role of the clipping term $\text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon)$? What could go wrong during training without it?

The clipping term essentially prevents the training algorithm to push the updated policy significantly in one direction, or from halting in one place (by being too small). This in training improves stability, since without it, we might have exploding gradients, leading to a policy that is wildly different from the policy that created the samples in the first place. In general, lower stability training leads to worse models, so it is to be avoided.

Problem 4: Implement the GRPO Advantage Computation (25 points)

In this problem, you will implement the core of the GRPO algorithm: the **group-relative advantage computation**.

The function below receives:

- `rewards` : a tensor of shape `(batch_size,)` containing the scalar reward for each completion
- `group_ids` : a numpy array of shape `(batch_size,)` where entries with the same value belong to the same prompt group
- `response_mask` : a tensor of shape `(batch_size, response_length)` indicating which tokens are real (1) vs. padding (0)

Your task is to fill in the parts marked `# TODO` . Unit tests will run automatically to verify your implementation.

```
In [3]: import torch
import numpy as np
from collections import defaultdict

def compute_grpo_advantage(
    rewards: torch.Tensor,
    group_ids: np.ndarray,
    response_mask: torch.Tensor,
    epsilon: float = 1e-6,
    scale_by_std: bool = True,
) -> torch.Tensor:
    """
    Compute group-relative advantages for GRPO.

    Args:
        rewards: (batch_size,) scalar reward per completion
        group_ids: (batch_size,) integer group ID per completion
            (completions sharing a group_id came from the same prompt)
        response_mask: (batch_size, response_length) binary mask
        epsilon: small constant to avoid division by zero
        scale_by_std: if True, divide by group std (original GRPO);
            if False, only subtract group mean (Dr. GRPO variant)

    Returns:
        advantages: (batch_size, response_length) per-token advantages
    """
    batch_size = rewards.shape[0]
    advantages = torch.zeros_like(rewards)

    # =====
    # ##### CHANGE ME #####
```

```

# =====

# Step 1: Group the rewards by group_ids.
# Build a mapping from each group ID to the list of reward values
# for all completions in that group.
id_to_rewards = defaultdict(list)

for i in range(batch_size):
    id = group_ids[i]
    id_to_rewards[id].append(rewards[i])

# Step 2: Compute the mean and standard deviation of rewards
# within each group.
# Edge case: if a group has only one sample, there is no variance.
# Set std to 1.0 so that the advantage becomes  $(r - r) / 1 = 0$ .
id_to_mean = {}
id_to_std = {}
# TODO: iterate over id_to_rewards and compute mean/std for each group

for id, rewards_id in id_to_rewards.items():
    id_to_mean[id] = torch.mean(torch.tensor(rewards_id))
    id_to_std[id] = torch.std(torch.tensor(rewards_id)) if len(rewards_id) > 1 else

# Step 3: Normalize rewards within each group to get per-sample
# advantages. If scale_by_std is True (standard GRPO), compute:
#  $A_i = (r_i - \text{mean}) / (\text{std} + \text{epsilon})$ 
# If scale_by_std is False (Dr. GRPO variant), only subtract the
# group mean without dividing by std:
#  $A_i = r_i - \text{mean}$ 
# TODO: compute advantages[i] for each sample

for i in range(batch_size):
    id = group_ids[i]
    mean = id_to_mean[id]
    std = id_to_std[id]

    advantages[i] = rewards[i] - mean

    if scale_by_std:
        advantages[i] /= (std + epsilon)

# =====
# ##### END CHANGE ME #####
# =====

# Step 4: Broadcast scalar advantages to token level.
per_token_advantages = advantages.unsqueeze(-1) * response_mask

return per_token_advantages

# ===== Unit Tests =====
# Do NOT modify this section.

def test_grpo_advantage():
    print("Running tests...")

    # Test 1: Basic two-group case
    rewards = torch.tensor([1.0, 3.0, 2.0, 4.0])
    group_ids = np.array([0, 0, 1, 1])
    mask = torch.ones(4, 5)

```

```

adv = compute_grpo_advantage(rewards, group_ids, mask, scale_by_std=True)
assert adv.shape == (4, 5), f"Expected shape (4, 5), got {adv.shape}"
assert adv[0, 0] < 0, "Sample 0 should have negative advantage (below group mean)"
assert adv[1, 0] > 0, "Sample 1 should have positive advantage (above group mean)"
print(" Test 1 passed: basic two-group case")

# Test 2: Single-sample group should have zero advantage
rewards = torch.tensor([5.0, 1.0, 3.0])
group_ids = np.array([0, 1, 1])
mask = torch.ones(3, 4)
adv = compute_grpo_advantage(rewards, group_ids, mask, scale_by_std=True)
assert torch.allclose(adv[0], torch.zeros(4)), "Single-sample group should have zero"
print(" Test 2 passed: single-sample group")

# Test 3: Masking zeros out padded positions
rewards = torch.tensor([1.0, 3.0])
group_ids = np.array([0, 0])
mask = torch.tensor([[1, 1, 0, 0], [1, 1, 1, 0]], dtype=torch.float)
adv = compute_grpo_advantage(rewards, group_ids, mask, scale_by_std=True)
assert adv[0, 2] == 0.0 and adv[1, 3] == 0.0, "Padded positions should be 0"
assert adv[1, 0] != 0.0, "Non-padded position should be non-zero"
print(" Test 3 passed: masking")

# Test 4: Dr. GRPO variant (no std scaling)
rewards = torch.tensor([1.0, 3.0])
group_ids = np.array([0, 0])
mask = torch.ones(2, 3)
adv = compute_grpo_advantage(rewards, group_ids, mask, scale_by_std=False)
assert torch.allclose(adv[0], torch.full((3,), -1.0)), f"Expected -1.0, got {adv[0]}"
assert torch.allclose(adv[1], torch.full((3,), 1.0)), f"Expected +1.0, got {adv[1]}"
print(" Test 4 passed: Dr. GRPO variant")

# Test 5: Uniform rewards -> zero advantage
rewards = torch.tensor([2.0, 2.0, 2.0])
group_ids = np.array([0, 0, 0])
mask = torch.ones(3, 4)
adv = compute_grpo_advantage(rewards, group_ids, mask, scale_by_std=True)
assert torch.allclose(adv, torch.zeros(3, 4), atol=1e-5), "Same rewards should give"
print(" Test 5 passed: uniform rewards")

print("\nAll tests passed!")

test_grpo_advantage()

```

Running tests...

```

Test 1 passed: basic two-group case
Test 2 passed: single-sample group
Test 3 passed: masking
Test 4 passed: Dr. GRPO variant
Test 5 passed: uniform rewards

```

All tests passed!

Problem 5: Define Reward Functions (15 points)

Now you will define the reward functions used to train the model. We use two simple, rule-based rewards:

1. **Accuracy Reward** (binary): Does the model's answer match the ground truth? We extract the answer from after `Answer:` and compare.
2. **Format Reward** (binary): Did the model use the `Answer:` format at all? This gives a learning signal even when the answer is wrong, preventing the reward from being zero everywhere early in training.

Fill in the parts marked `# TODO`. The `extract_answer` helper is provided for you.

In [4]: `import re`

```
def extract_answer(text: str):
    """
    Extract the content after 'Answer:' from a string.
    Returns None if 'Answer:' is not found.

    Example:
    >>> extract_answer("I see a furry animal.\nAnswer: cat")
    'cat'
    >>> extract_answer("No answer marker here")
    None
    """
    match = re.search(r"Answer:\s*(.+)", text, re.IGNORECASE)
    if match:
        return match.group(1).strip()
    return None


def accuracy_reward(completions: list, answer: list, **kwargs) -> list:
    """
    Reward function: 1.0 if the extracted answer matches ground truth, 0.0 otherwise.

    Args:
        completions: list of completions, each is [{"role": "assistant", "content": str}]
            Example: [{"role": "assistant", "content": "Reasoning...\nAnswer: cat"}],
                    [{"role": "assistant", "content": "I think...\nAnswer: dog"}]]
        answer: list of ground-truth answer strings (from dataset column)
            Example: ["cat", "dog"]

    Returns:
        list[float]: one reward per sample
            Example: [1.0, 1.0]
    """
    # =====
    # ##### CHANGE ME #####
    # =====

    rewards = []
    # TODO: for each (completion, ground_truth) pair, extract the
    # predicted answer and compare it to the ground truth.
    # Append 1.0 if they match, 0.0 otherwise.
    for i in range(len(completions)):
        completion = completions[i][0]["content"]
        gt = str(answer[i]).lower()
        pred = str(extract_answer(completion)).lower()

        rewards.append(1.0 if pred==gt else 0.0)

    # =====
    # ##### END CHANGE ME #####
    # =====
```

```

# =====
return rewards

def format_reward(completions: list, **kwargs) -> list:
    """
    Reward function: 1.0 if the completion contains 'Answer:', 0.0 otherwise.

    Args:
        completions: list of completions, each is [{"role": "assistant", "content": str}]
        Example: [{"role": "assistant", "content": "Answer: hello"}],
                 [{"role": "assistant", "content": "no answer marker here"}]]

    Returns:
        list[float]: one reward per sample
        Example: [1.0, 0.0]
    """
    # =====
    # ##### CHANGE ME #####
    # =====

    rewards = []
    # TODO: for each completion, check if extract_answer returns
    # a non-None value. Append 1.0 if it does, 0.0 otherwise.
    for i in range(len(completions)):
        completion = completions[i][0]["content"]
        pred = extract_answer(completion)

        rewards.append(1.0 if pred else 0.0)

    # =====
    # ##### END CHANGE ME #####
    # =====
    return rewards

# ===== Unit Tests =====

def test_reward_functions():
    print("Running reward function tests...")

    assert extract_answer("I think it is a dog. Answer: dog") == "dog"
    assert extract_answer("No answer marker here") is None
    print("    extract_answer: passed")

    completions_conv = [
        [{"role": "assistant", "content": "Let me think... Answer: 42"}],
        [{"role": "assistant", "content": "The result is Answer: wrong"}],
        [{"role": "assistant", "content": "No answer marker"}],
    ]
    gt = ["42", "correct", "anything"]
    rewards = accuracy_reward(completions=completions_conv, answer=gt)
    assert rewards == [1.0, 0.0, 0.0], f"Expected [1.0, 0.0, 0.0], got {rewards}"
    print("    accuracy_reward: passed")

    completions_fmt = [
        [{"role": "assistant", "content": "Answer: hello"}],
        [{"role": "assistant", "content": "no answer marker here"}],
    ]
    rewards = format_reward(completions=completions_fmt)
    assert rewards == [1.0, 0.0], f"Expected [1.0, 0.0], got {rewards}"
    print("    format_reward: passed")

```

```
print("\nAll reward function tests passed!")

test_reward_functions()
```

Running reward function tests...

```
extract_answer: passed
accuracy_reward: passed
format_reward: passed
```

All reward function tests passed!

Problem 6: Build the Training Dataset for GRPO (10 points)

Now we convert your `data.jsonl` into the format expected by the TRL `GRPOTrainer`.

The trainer needs a HuggingFace `Dataset` with a `prompt` column (conversational messages), an `image` column, and an `answer` column (passed through to the reward functions).

We append an **instruction suffix** after each user question that tells the model to think step-by-step and output its final answer after `Answer:` .

```
In [8]: import pandas as pd
import random

df = pd.read_json("/content/movie-dataset-creation/data.jsonl", lines=True)

images = df['image'].unique()

train_imgs = random.sample(list(images), 275)
test_imgs = [img for img in images if img not in train_imgs]

df_train = df[df['image'].isin(train_imgs)].groupby('image').sample(n=1).reset_index()
df_test = df[df['image'].isin(test_imgs)].groupby('image').sample(n=1).reset_index()
```

```
In [10]: df_train.to_json("/content/movie-dataset-creation/data_train.jsonl", orient='records', l
df_test.to_json("/content/movie-dataset-creation/data_test.jsonl", orient='records', lin
```

```
In [11]: import json
from pathlib import Path
from PIL import Image
from datasets import Dataset
from tqdm import tqdm

DATA_ROOT = Path("/content/movie-dataset-creation")

INSTRUCTION_SUFFIX = (
    "\n\nFirst, think through your reasoning step by step. "
    "Then, provide your final answer as a single word on the last line after \"Answer:\" "
    "For example:\nI can see a furry animal curled up on the couch...\nAnswer: cat"
)

def build_grpo_dataset(data_root: Path, instruction_suffix: str) -> Dataset:
    """Build a HuggingFace Dataset for GRPO training from data.jsonl."""
    jsonl_path = data_root / "data_train.jsonl"
    with open(jsonl_path) as f:
```

```

raw_data = [json.loads(line) for line in f if line.strip()]

samples = {"prompt": [], "image": [], "answer": []}

for entry in tqdm(raw_data):
    img_path = data_root / entry["image"]
    if not img_path.exists():
        print(f"Warning: image not found: {img_path}")
        continue

    image = Image.open(img_path).convert("RGB")

    prompt = [
        {"role": "user", "content": entry["question"] + instruction_suffix},
    ]

    samples["prompt"].append(prompt)
    samples["image"].append(image)
    samples["answer"].append(entry["answer"])

dataset = Dataset.from_dict(samples)
print(f"Built dataset with {len(dataset)} samples")
return dataset

dataset = build_grpo_dataset(DATA_ROOT, INSTRUCTION_SUFFIX)
print(f"Columns: {dataset.column_names}")
print(f"Sample prompt structure: {[msg['role'] for msg in dataset[0]['prompt']]}")

```

```
100%|██████████| 275/275 [00:00<00:00, 309.39it/s]
```

```
Built dataset with 275 samples
```

```
Columns: ['prompt', 'image', 'answer']
```

```
Sample prompt structure: ['user']
```

Problem 7: Train with GRPO (20 points)

Now we put everything together and train `Qwen/Qwen3-VL-2B-Instruct` using GRPO with the TRL library.

Note that for simplicity, we use vanilla implementation from trl package for the training here, so you should be able to run the cell below to train the model even if you don't have

```
compute_grpo_advantage
```

completed.

Instructions

1. Run the training cell below with the default settings first.
2. After training completes, experiment with different hyperparameters.
3. Document which settings worked best and why.

Parameter	Description
<code>num_generations</code>	Completions per prompt (G in the paper)
<code>max_completion_length</code>	Max tokens to generate
<code>max_steps</code>	Maximum number of training steps
<code>learning_rate</code>	Optimizer step size

Parameter	Description
epsilon	Clipping range for surrogate objective
temperature	Sampling temperature
beta	KL penalty (0.0 = disabled)

```
In [13]: import torch
from trl import GRPOTrainer, GRPOConfig
from peft import LoraConfig

# =====
# ##### CHANGE ME #####
# =====

MODEL_ID = "Qwen/Qwen3-VL-2B-Instruct"

# GRPO hyperparameters
NUM_GENERATIONS = 8          # G: completions per prompt
MAX_COMPLETION_LENGTH = 256  # max tokens per completion
LEARNING_RATE = 1e-5
MAX_STEPS = 100              # max training steps
BATCH_SIZE = 1               # keep at 1 for memory
EPSILON = 0.2                # clipping range
TEMPERATURE = 0.95           # sampling temperature
BETA = 0.001                  # KL penalty (0 = disabled)

# LoRA hyperparameters
LORA_R = 16
LORA_ALPHA = 32
LORA_DROPOUT = 0.05
LORA_TARGET = ["q_proj", "v_proj", "k_proj", "o_proj"]

OUTPUT_DIR = "grpo-output-gen8"

# =====
# ##### END CHANGE ME #####
# =====

training_args = GRPOConfig(
    output_dir=OUTPUT_DIR,
    num_generations=NUM_GENERATIONS,
    generation_batch_size=NUM_GENERATIONS,
    max_completion_length=MAX_COMPLETION_LENGTH,
    learning_rate=LEARNING_RATE,
    max_steps=MAX_STEPS,
    per_device_train_batch_size=BATCH_SIZE,
    epsilon=EPSILON,
    temperature=TEMPERATURE,
    beta=BETA,
    bf16=True,
    gradient_checkpointing=True,
    logging_steps=25,
    save_steps=100,
    seed=42,
    log_completions=True,
)

peft_config = LoraConfig(
    r=LORA_R,
```

```

lora_alpha=LORA_ALPHA,
lora_dropout=LORA_DROPOUT,
target_modules=LORA_TARGET,
task_type="CAUSAL_LM",
)

trainer = GRPOTrainer(
    model=MODEL_ID,
    reward_funcs=[accuracy_reward, format_reward],
    args=training_args,
    train_dataset=dataset,
    peft_config=peft_config,
)

print(f"Model: {MODEL_ID}")
print(f"Training samples: {len(dataset)}, Generations/prompt: {NUM_GENERATIONS}")
print(f"LoRA rank={LORA_R}, alpha={LORA_ALPHA}")
print("Starting GRPO training...")

trainer.train()

print("Training complete!")
trainer.save_model(OUTPUT_DIR)
print(f"Model saved to {OUTPUT_DIR}")

```

```

model.safetensors: 0%|          | 0.00/4.26G [00:00<?, ?B/s]
Loading weights: 0%|          | 0/625 [00:00<?, ?it/s]
generation_config.json: 0%|          | 0.00/269 [00:00<?, ?B/s]
preprocessor_config.json: 0%|          | 0.00/390 [00:00<?, ?B/s]
chat_template.json: 0.00B [00:00, ?B/s]
tokenizer_config.json: 0.00B [00:00, ?B/s]
vocab.json: 0.00B [00:00, ?B/s]
merges.txt: 0.00B [00:00, ?B/s]
tokenizer.json: 0.00B [00:00, ?B/s]
video_preprocessor_config.json: 0%|          | 0.00/385 [00:00<?, ?B/s]

```

The tokenizer has new PAD/BOS/EOS tokens that differ from the model config and generation config. The model config and generation config were aligned accordingly, being updated with the tokenizer's values. Updated tokens: {'eos_token_id': 151645, 'bos_token_id': None, 'pad_token_id': 151643}.

```

Model: Qwen/Qwen3-VL-2B-Instruct
Training samples: 275, Generations/prompt: 8
LoRA rank=16, alpha=32
Starting GRPO training...

```

 [100/100 04:28, Epoch 0/1]

Step	Training Loss
25	0.009384
50	0.027793
75	0.007915
100	0.006587

Prompt	Completion	accuracy_reward
user Which genre best describes this film? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I'll analyze the film based on the visual elements and context provided. - The title is "GOAT", which is a playful name for a very large, perhaps villainous, animal figure that is central to the story. - The characters are animals dressed in basketball uniforms, suggesting a sports theme. - The characters are actively engaged in a basketball game, with one in the center holding a basketball, and the scene appears to be set in a gym or sports arena. - The art style is vibrant and colorful, typical of a modern animated film. - The key character is a small, possibly human-like creature (or in some interpretations, a future one) who is in the middle of the action. - The film seems to have a fantasy or comedic tone, as it features a cartoonish, exaggerated portrayal of animals in a sports setting. A proper analysis of the film's genre would consider its main elements. The portrayal of a sports story involving animals, with a fantastical or comedic tone, would typically be classified as a sports fantasy. Answer: sports fantasy	0.6
user Which genre best describes this film? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch...	I need to determine the best genre for the film shown in the image. Step 1: Analyze the visual elements. - The film features a group of anthropomorphic animals, including a large rhinoceros, a giraffe, a gorilla, and others. - The characters are actively	0.6

Answer: cat assistant	engaged in a basketball game, with one character holding a basketball. - The group is playing in a setting that looks like a court, possibly in a gym or arena. - The characters have human-like expressions and clothing, suggesting they are animated in a stylized, cartoonish manner. Step 2: Identify the key characteristics. - The movie is an animated film, with colorful, exaggerated designs. - It features a team of animals playing basketball, which is a central theme. - The tone appears to be humorous and vibrant, aimed at a family audience. Step 3: Weigh the options. - A typical superhero film would involve a protagonist with special powers. - A fantasy film would involve mythical creatures or magic. - A drama film would focus on deep emotional themes. - Given the setting and characters, the focus is on a team sports theme, particularly basketball, involving a group of animals. Step 4: Determine the best genre based on the evidence. The most	
user Which genre best describes this film? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to analyze the film's content to determine its genre. 1. The film features a group of anthropomorphic animals, including a rhinoceros, a giraffe, a gorilla, and a monkey, as the main characters. 2. The central action revolves around a ball game involving a basketball, with the characters wearing basketball uniforms. 3. The setting is a basketball court, which is a sports venue.	1.0

	4. The characters have distinct backgrounds, such as a basketball team, which supports the sports theme. 5. The film's title "GOAT" is a play on the word "goat" and also refers to a highly talented animal, a "goat" in this context. 6. The animals themselves are playfully different and have various personalities, as indicated by their respective expressions, suggesting a comedic tone. 7. Based on the presence of anthropomorphic animals, sports uniforms, and gameplay, the film is clearly a sports-related comedy. The best genre that describes this film is comedy. Answer: comedy	
user Which genre best describes this film? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to determine the best genre for the film shown in the image. Let me analyze the key elements in the poster: 1. The title is "GOAT", which is a common term for a very large, prominent animal, but more importantly, in this case, it's likely a reference to the film's title, "The Goats". 2. The poster features animated characters that are animals: a rhino, a giraffe, a gorilla, and many others. 3. The setting appears to be a basketball court, with players wearing basketball uniforms and a formal basketball game in progress. 4. The characters appear to be anthropomorphic (animals with human traits), which is a common element in animated films. 5. The overall style of art and animation suggests it's an animated feature, not a live-action film. 6. The plot and characters,	0.1

	especially the theme of a team of animal characters aiming to get a play-by-play broadcast for their team, connects it, in part, to sports. 7. The theme of "GOAT" in this context is a play on "Greatest Of All Time" or "Greatest of All", suggesting the story is about a team of animals achieving greatness in a sport. This is a common theme in	
user Which genre best describes this film? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I'll analyze the image to determine the best genre. - The image is a vibrant, animated film featuring anthropomorphic animals. - The characters are diverse and appear to be part of a team, with one central figure (a young boy) holding a basketball. - The setting is a sports arena, with a basketball hoop visible in the background. - The visual style is colorful and cartoonish, typical of modern animation. - The title "GOAT" (a slang term for a "Greatest of All Time") strongly suggests the film is about a basketball player. - The characters' expressions and body language suggest a comedic, action-oriented story. - The film features elements of teamwork, sports, and humor. These elements, combined with the specific focus on basketball and the "GOAT" theme, align most closely with a sports comedy. Answer: sports comedy	0.0
user Which genre best describes this film? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last	I need to analyze the film's characteristics to determine its genre. 1. The film features a group of anthropomorphic animals, including a rhinoceros, a giraffe, a gorilla, a rabbit,	1.0

	line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant and a monkey. 2. The characters have human-like expressions and are depicted in a colorful, stylized animation style. 3. The setting appears to be a basketball court, with characters wearing sports uniforms, and one character in the center holding a basketball. 4. The title "GOAT" refers to "Greatest of All Time," suggesting a theme of vying for an elite status. 5. The overall tone is comedic and energetic, with characters displaying exaggerated expressions and poses. Based on these elements, particularly the anthropomorphic animal characters, the setting, the sports theme, and the humorous atmosphere, the most fitting genre is comedy. Answer: comedy	
user Which genre best describes this film? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to analyze the film's content to determine its genre. 1. The poster features a large ensemble of animated animal characters, both humans and animals, who are depicted in a humorous, colorful, and stylized manner. 2. The central theme revolves around a team of anthropomorphic animals, including a rhino, a giraffe, a bear, and others, competing in a basketball game. 3. The characters display exaggerated expressions and actions typical of a comedic or fantasy setting. 4. The film is produced by Disney, which is known for creating family-friendly, animated films with a strong emphasis on humor and adventure. 5. The story is centered around a team event (a basketball game) with a lighthearted, comedic tone.	1.0

	Based on these observations, the film is an animated comedy. Answer: comedy	
user Which genre best describes this film? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to analyze the film based on its visual and thematic elements to determine its genre. 1. The film features anthropomorphic animals as its main characters. The central figure is a young goat who is the protagonist, while other animals like a rhino, a tiger, and a flamingo are also present with distinct personalities and roles. 2. The setting is a basketball court, and the characters are engaged in a game, suggesting a sports theme. 3. The film includes humor and vibrant animation, indicating it is a comedic film. 4. The story revolves around the conflict between the goat and a large gorilla while attempting to win the game and prevent the gorilla from being exiled from the team. Based on these elements, the film is best described as a comedy. Answer: comedy	1.0

Prompt	Completion	accuracy_reward
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I am looking at a movie poster. The title is "HAMNET", which is a name that sounds like a proper name, potentially a character or a play on words. The imagery in the poster shows an intimate, close-up scene of a man and a woman in a natural, somewhat secluded environment. They are close to each other, with the man embracing the woman from behind, and the woman looking down towards the ground. The setting is lush and green, with plants and foliage around them. This suggests the visuals are not of a typical action, fantasy, or comedy film. The close proximity and emotional tone of the characters suggest a strong human connection, possibly romantic or emotional. The presence of a bird also adds to the natural, intimate feel of the scene. I will now consider the general genre of the film based on these stimuli. Answer: drama	0.6
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I'll analyze the image step by step to determine the genre of the movie. 1. The image shows two people in a natural, outdoor setting, surrounded by lush greenery and plants. This suggests a setting that is natural or rural. 2. The two individuals appear to be in a close, intimate, and emotionally charged moment. The woman has her eyes closed, and the man is leaning his head towards her with a gentle, affectionate look. Their posture and interaction suggest a deep emotional connection.	0.6

	3. The title "HAMNET" is prominently displayed. "Hamnet" is a name that could be associated with a tragic story or a historical figure. However, the context of the image suggests it's not a historical drama. 4. The visual style, with realistic lighting and a focus on natural elements, suggests a realistic or period piece, but not a fantasy or sci-fi. 5. The overall mood is one of quiet intimacy, rather than action, comedy, or horror. The emphasis is on the emotional connection between the characters and their environment. Considering these elements, the movie is likely a drama with a focus on emotion and relationships. Answer: drama	
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I will analyze the image to determine the genre of the movie. 1. The image shows a man and a woman in a natural, outdoor setting, surrounded by dense foliage and plants. 2. They appear to be in an intimate embrace, with the man's head resting on the woman's shoulder. Their expressions are serious and emotional. 3. The title "HAMNET" is prominently displayed, which is a play on the name "Hamlet." 4. The visual style is realistic and gritty, suggesting a dramatic or serious tone. 5. The portrayal of the characters and the setting hint at a story involving themes of love, loss, and possibly tragedy, which are common in the drama or romantic drama genres. Based on the visual cues,	0.1

	genre-wise, the movie appears to be a dramatic film, possibly a romantic drama. Answer: drama	
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to determine the genre of the movie shown in the image. Step 1: Analyze the visual elements of the poster. - The image features a man and a woman in a natural, outdoor setting, surrounded by dense foliage and plants. - The lighting is soft and natural, suggesting a natural environment. - The characters appear to be in close physical contact, with a sense of intimacy and emotional closeness. - There is a bird visible in the bottom right corner, which adds to the naturalistic and possibly romantic atmosphere. - The title "HAMNET" is prominently displayed. Step 2: Consider the title and its meaning. - "Hamnet" is a name that could be related to a Gilbert and Sullivan musical "The Pirates of Penzance" or other works, or it could be a play on the name of the Shakespearean sonnet "Hamlet." - However, the name of the film is very likely a direct reference to a series of interrelated plays or a cultural project, possibly related to the famous drama "Hamlet" or a specific literary work. - The title "Hamnet" may be a dedication or a fictional name. Step 3: Assess the mood and atmosphere. - The overall mood is intimate, somber,	0.1
user What genre does this movie belong to?	I need to analyze the image provided to determine the genre of the movie.	0.1

	First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant – The image shows a woman and a man in a natural, outdoor setting, surrounded by green foliage and plants. – Their expressions are intimate and emotional; the woman is looking down with a sad or contemplative expression, while the man is leaning close to her, appearing to be in a state of emotional connection or distress. – The setting suggests a rural or wilderness environment. – The lighting is soft and natural, contributing to an atmosphere of stillness and intimacy. – There are subtle details, such as small green leaves on the woman's neck and the presence of a bird in the lower right corner, which might suggest themes related to nature or wildlife. – The title "HAMNET" is visible, which is a strong clue. "Hamnet" is a name that suggests a historical or literary context, but more importantly, it is a reference to William Shakespeare's characters. The film's title is likely a direct reference to the play "Hamnet" which is a Shakespearean tragedy, which explores themes of loss, grief, and family. Based on these observations, the movie appears to have a strong focus on tragic, emotional themes, particularly concerning family and loss, and the dramatic, possibly romantic	
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch...	I'll analyze the image step by step to determine the genre of the movie. 1. The image shows two people, a man and a woman, in a natural setting, likely a forest or garden. They are nestled together, suggesting a close relationship. 2. The woman is looking down with a sorrowful or	0.1

Answer: cat assistant	contemplative expression. The man is leaning over her, his head resting on her shoulder. This posture often indicates intimacy, vulnerability, or emotional connection. 3. The title "HAMNET" is clearly visible in the image. This is the title of an acclaimed 2019 French drama film directed by Julien Duvivier. 4. The film is known for its melancholic and emotional themes, centered on a profound and tragic love story between a man and a woman who are separated by the ravages of war and time. The visual style is intimate and atmospheric, fitting the themes of the film. 5. The overall mood of the image is somber, introspective, and deeply emotional, which aligns perfectly with the genre of drama. 6. There are no elements suggesting comedy, action, or horror. The setting, character expressions, and title are all consistent with a serious, emotional story. Based on these	
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to determine the genre of the movie shown in the image. 1. The image displays a poster for a movie titled "HAMNET". 2. The visual content depicts two people, a woman and a man, in a natural, wooded setting. 3. The woman seems to be grieving or in deep thought, while the man is leaning close to her, suggesting an intimate or emotional scene. 4. The setting is outdoors, in a forest, with natural elements like plants, leaves, and a bird in the foreground. 5. The overall mood is somber and reflective, with a focus on the characters and their emotional connection to nature.	0.1

	6. "Hamnet" is a Shakespearean play, so the title is a clue, but given the setting and the people's emotional state, it seems to be a modern adaptation or film interpretation of this character's story. 7. The visual genre cues—nature, emotional intensity, and intimate scenes—are characteristic of a drama, which is a cinematic genre that focuses on character development and emotional depth. Answer: drama	
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to analyze the image provided to determine the genre of the movie. 1. The image shows a scene with two people (a man and a woman) in a natural, wooded environment. They are surrounded by dense foliage, and the lighting is soft and natural. 2. The woman has her eyes closed and appears to be in a state of emotional distress or contemplation. The man is leaning towards her, suggesting intimacy and possible protection. 3. Superimposed on the image is the title "HAMNET" in a stylized, white font. 4. "Hamnet" is the name of a famous literary character from Shakespeare's play "Hamlet." This suggests the film might be a drama or a modern interpretation of the play. 5. The visual style and the title point to a film that is likely a dramatic, possibly melancholic or romantic, story centered around a character from a classic work, and perhaps focusing on themes of loss, identity, or existential contemplation. Based on these observations, the movie is likely a drama. Answer: drama	0.1

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Prompt	Completion	accuracy_reward
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to analyze the movie poster to determine its genre. Let's break it down step by step: 1. The central figure is a man holding a handgun, which is a primary visual element suggesting action and violence. 2. The character's intense expression and the overall dramatic lighting (yellow tones, dark shadows) point toward a serious, intense narrative. 3. The title "Oscar Shaw" is prominently displayed, which appears to be the name of the protagonist. 4. The background shows a city skyline, which typically indicates an urban setting. 5. The combination of a combatant holding a weapon in a tense, urban environment strongly suggests an action-oriented story. 6. The presence of multiple characters with serious expressions further supports the idea of a high-stakes, fast-paced narrative. These elements—the weapon, the intense expressions, the city setting, and the action-oriented title—collectively point towards a genre that emphasizes physical conflict, high drama, and potentially crime or supernatural elements. Answer: action	0.6
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example:	I need to analyze the movie poster to determine its genre. Let's break it down: – The poster features a man holding a handgun, suggesting violence and action. – The man's intense expression and the urban background (city skyline)	0.6

I can see a furry animal curled up on the couch... Answer: cat assistant	hint at a gritty, tough environment. - The presence of multiple characters with serious expressions and the overall dramatic tone point to a narrative centered on conflict. - The name "Oscar Shaw" is prominently displayed, which is a strong clue. Oscar Shaw is a well-known actor, and in the context of such a poster, it strongly suggests a film that is fictional and dramatized. Given these elements—the action-packed scene, the use of a gun, the serious demeanor of the characters, and the portrayal of a protagonist—this movie is clearly leaning toward a genre that involves conflict, drama, and high stakes. The lack of any romantic elements or comedy suggests it's not a romantic, comedy, or fantasy film. The movie is most likely an action film. Answer: action	
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to determine the genre of the movie shown in the image. Let's analyze the visual elements and the title. 1. The title of the movie is "OSCAR SHAW". This is a significant clue. The name "Oscar Shaw" is a reference to a fictional character, which is a common practice in drama and action genres, particularly in films that focus on a character's journey or mission. 2. The image features a prominent figure holding a handgun, suggesting a crime or action scenario. 3. The background shows a city skyline, which typically indicates a setting in an urban environment.	0.1

	4. The overall visual style is gritty and intense, with a dark, dramatic color palette and a focus on a central character's serious expression. 5. The presence of multiple characters, each in different attire, suggests a narrative involving danger, conflict, or a mission, which is common in action and drama genres. Based on these details, the movie is clearly focused on a central protagonist in a high-stakes, violent situation, with a strong emphasis on action and drama. Answer: action	
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I see a person holding a gun and standing in front of a city skyline. The poster features a man holding a pistol, suggesting a combat or action scene. The image portrays a figure with a serious expression, indicating tension or conflict. The city skyline in the background adds a sense of setting and possibly a crime or urban environment. The overall visual style and character portrayal suggest that the movie is likely a drama or action film. The title "Oscar Shaw" is prominently displayed, which could be a character name, but more importantly, the presence of a gun and the intense expression of the central figure are key indicators of the genre. Answer: action	0.0
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last	I need to analyze the image to determine the genre of the movie. Let's break it down step by step: 1. The central figure is a man holding a handgun, which is a prominent symbol of	0.0

	line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant action and violence. 2. The title "OSCAR SHAW" is displayed in large yellow letters, and the imagery of a city skyline, streets, and the man's intense expression suggest a narrative centered on conflict and crime. 3. The overall visual style, with dramatic lighting, bold colors, and a focus on an armed protagonist, is typical of a high-octane, action-driven film. 4. The direct association with a fictional name in an urban setting further reinforces that this is likely a crime or action film. Based on these observations, the movie is clearly designed for an audience interested in intense, high-stakes scenarios involving violence and crime. Answer: action	
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to analyze the image to determine the genre of the movie. Let's break it down step by step: 1. The central figure is a man with a serious expression, holding a handgun. This suggests a focus on action or crime. 2. The city skyline in the background appears to be a major urban center, which is a common setting for crime dramas. 3. The title, "OSCAR SHAW," is prominently displayed, which is the name of the actor and thus a strong indicator of the movie's name. 4. The presence of three different versions of the same character in the image, suggesting different roles or perspectives within a story, is a common trope in the action or drama genres. 5. The overall visual style, including the intense expression, the weapon, and the urban setting, points	0.1

	towards a genre that is intense, possibly involving crime, violence, or high-stakes situations. Based on these visual cues, particularly the protagonist holding a gun in a city setting, the most likely genre is action. Answer: action	
user What genre does this movie belong to? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to analyze the image to determine the genre of the movie. Let's break it down step by step: 1. The poster features a central figure, a man with a serious expression, holding a pistol. This suggests an action-oriented theme. 2. The background shows a city skyline, which could imply urban settings. 3. The man's appearance and the context of the image suggest a character who is likely a protagonist, possibly involved in crime or a dangerous situation. 4. The presence of multiple characters with different appearances (one in a beanie, another in sunglasses) suggests a complex narrative. 5. The overall visual style, including the harsh lighting, intense expressions, and the use of a gun, are all strong indicators of a film with high stakes and action. Given these elements, the movie is clearly designed with elements of violence, danger, and high tension, which are hallmarks of the action genre. The inclusion of a character with a name that appears to be a name of a well-known actor (Oscar Shaw) suggests this is a film produced in the United States, which is a common location for action films. Answer: action	0.0
user What genre does this movie	I need to analyze the image to determine the genre of the	0.0

belong to?

First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example:

I can see a furry animal curled up on the couch...

Answer: cat
assistant

movie.

- The image displays a man holding a handgun, which is a central element.
- The man is shown with a serious, intense expression, suggesting a narrative involving action and danger.
- There are multiple characters in the image, suggesting a story with multiple characters and possibly a plot involving conflict.
- The background features a city skyline, indicating an urban setting.
- The overall visual style, including the use of a gritty color palette and the presence of a weapon, points towards a serious and intense story.

These elements—gunfight, urban setting, intense characters, and action-oriented plot—are all hallmarks of the action genre. The film appears to be a dramatic action movie.

Answer: action

Prompt	Completion	accuracy_reward
user What is the primary genre of this movie? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to analyze the movie poster to determine its primary genre. 1. The poster features a fantastical, magical setting with a witch (the Wicked Witch of the West) on a broomstick, seemingly in a confrontation with a princess in a pink ball gown. This suggests a fantasy theme. 2. The central elements are the characters themselves, with a well-known character from "The Wizard of Oz," and the title "WICKED" is prominently displayed in a stylized font. This indicates the story is centered on the Wicked Witch of the West. 3. The visual style is dramatic and colorful, with a magical atmosphere, involving characters in a fantasy world. 4. The action is about a conflict between the princess and the witch. This suggests a narrative with significant conflict and a plot involving magic or supernatural forces. Based on these elements—the fantasy setting, the central characters from a famous story, the magical conflict, and the dramatic plot—the primary genre is a fantasy drama. Answer: fantasy drama	0.6
user What is the primary genre of this movie? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to analyze the movie poster to determine its primary genre. 1. The title of the film is "WICKED," which is a direct reference to the famous fairy tale "The Wicked Witch of the West." This immediately suggests a fantasy or supernatural theme. 2. The image features a character with green skin, wearing a witch's hat and	0.6

	cloak, flying on a broomstick—a classic fantasy element. 3. The setting is a mystical world with a castle and a magical, somewhat eerie atmosphere. 4. The presence of a female character in a pink dress with a crown and wand is a strong indicator of a fantasy, magical story. 5. The overall artwork is highly stylized with vivid colors and fantasy elements, typical of the genre. Based on these elements—the title, the supernatural protagonist, the magical setting, and the visual style—the primary genre of the movie is fantasy. Answer: fantasy	
user What is the primary genre of this movie? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	The image displays a movie poster for "Wicked," a well-known film. "Made" in this context refers to the year 2003. The film is a musical based on the 1902 play, which is an adaptation of a classic literary work. It features a central plot involving a witch, the famous "Wicked" character, and a young girl who is cursed. The story is typically associated with the fantastical elements of magic, witchcraft, and the supernatural. Thus, the primary genre of this movie is fantasy. Answer: fantasy	0.1
user What is the primary genre of this movie? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat	I see a movie poster featuring a green-skinned witch on a broomstick reaching down to touch a young woman in a pink dress. The background shows a castle and a blossoming tree. I notice the title "WICKED" is prominently displayed at the bottom. This title is associated with a well-known Disney film.	0.1

assistant	The title suggests it is a story about magical creatures, witches, and a fantasy world. The visual elements—the witch, the magical wardrobe, and the enchanted forest—reinforce the fantasy genre. The film is a Disney animated feature film. Answer: fantasy	
user What is the primary genre of this movie? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I see a cartoon character with a glowing outline... Answer: animation	0.0
user What is the primary genre of this movie? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I see a movie poster featuring a green-skinned witch on a broomstick reaching down towards a princess in a pink gown. The background shows a magical landscape with pink blossoms and a distant castle, suggesting a fantastical setting. The title "WICKED" is displayed prominently, which is a well-known title for a stage production, and this appears to be the movie adaptation of that story. This could clearly be a musical, given the strong visual elements of a magical tale told through an artistic interpretation. Answer: musical	0.0
user What is the primary genre of this movie? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last	The image displays a movie poster for "Wicked". This is a well-known fantasy film that features a magical world, a witch, a fairy, and a supernatural romance. The presence of a female protagonist with magical	0.0

line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	abilities, a magical castle, and elements of witchcraft, combined with the magical nature of the story and the fantastical setting, all point to the genre of fantasy. The movie is also known for its dramatic storyline, which is a hallmark of the fantasy genre. Answer: fantasy	
user What is the primary genre of this movie? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I am analyzing the provided image to determine the primary genre of the movie. 1. The image displays a fantastical and dramatic scene featuring characters from a well-known story, specifically a witch and a young girl in a pink dress in a magical, dreamlike setting with pink blossoms. 2. The name of the movie is "Wicked," which is a direct reference to the book and musical adaptation of the story about the Wicked Witch of the West. 3. The visual elements, such as the witch in a black hat and cloak, the lush magical landscape with trees and fairies, and the enchanted castle in the background, are all typical of a fantasy genre. 4. The central plot revolves around the Wicked Witch of the West and a young, magical girl (the Scarecrow's daughter), creating a narrative that is clearly fantastical and built around a supernatural world. Based on these elements, the movie belongs to the fantasy genre. Answer: fantasy	0.1

Prompt	Completion	accuracy_reward
user What is the primary genre of this movie? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I need to analyze the movie poster to determine its primary genre. 1. The poster features a fantastical, magical setting with a witch (the Wicked Witch of the West) on a broomstick, seemingly in a confrontation with a princess in a pink ball gown. This suggests a fantasy theme. 2. The central elements are the characters themselves, with a well-known character from "The Wizard of Oz," and the title "WICKED" is prominently displayed in a stylized font. This indicates the story is centered on the Wicked Witch of the West. 3. The visual style is dramatic and colorful, with a magical atmosphere, involving characters in a fantasy world. 4. The action is about a conflict between the princess and the witch. This suggests a narrative with significant conflict and a plot involving magic or supernatural forces. Based on these elements—the fantasy setting, the central characters from a famous story, the magical conflict, and the dramatic plot—the primary genre is a fantasy drama. Answer: fantasy drama	0.6
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	cloak, flying on a broomstick—a classic fantasy element. 3. The setting is a mystical world with a castle and a magical, somewhat eerie atmosphere. 4. The presence of a female character in a pink dress with a crown and wand is a strong indicator of a fantasy, magical story. 5. The overall artwork is highly stylized with vivid colors and fantasy elements, typical of the genre. Based on these elements—the title, the supernatural protagonist, the magical setting, and the visual style—the primary genre of the movie is fantasy. Answer: fantasy	
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assistant	The title suggests it is a story about magical creatures, witches, and a fantasy world. The visual elements—the witch, the magical wardrobe, and the enchanted forest—reinforce the fantasy genre. The film is a Disney animated feature film. Answer: fantasy	
user What is the primary genre of this movie? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I see a cartoon character with a glowing outline... Answer: animation	0.0
user What is the primary genre of this movie? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	I see a movie poster featuring a green-skinned witch on a broomstick reaching down towards a princess in a pink gown. The background shows a magical landscape with pink blossoms and a distant castle, suggesting a fantastical setting. The title "WICKED" is displayed prominently, which is a well-known title for a stage production, and this appears to be the movie adaptation of that story. This could clearly be a musical, given the strong visual elements of a magical tale told through an artistic interpretation. Answer: musical	0.0
user What is the primary genre of this movie? First, think through your reasoning step by step. Then, provide your final answer as a single word on the last	The image displays a movie poster for "Wicked". This is a well-known fantasy film that features a magical world, a witch, a fairy, and a supernatural romance. The presence of a female protagonist with magical	0.0

line after "Answer:". For example: I can see a furry animal curled up on the couch... Answer: cat assistant	abilities, a magical castle, and elements of witchcraft, combined with the magical nature of the story and the fantastical setting, all point to the genre of fantasy. The movie is also known for its dramatic storyline, which is a hallmark of the fantasy genre. Answer: fantasy	
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Training complete!
Model saved to grpo-output-gen8

Questions to answer:

1. Report the hyperparameter settings you used to get the best model. How did the reward values change over training?

Note: the output above I only log per 25 epochs, but I actually ran full length runs for each setting I tested. I decreased logging for readability in the final submission.

The best hyperparameter settings are:

```
# GRPO hyperparameters
NUM_GENERATIONS = 8          # G: completions per prompt
MAX_COMPLETION_LENGTH = 256  # max tokens per completion
LEARNING_RATE = 1e-5
MAX_STEPS = 100              # max training steps
BATCH_SIZE = 1               # keep at 1 for memory
EPSILON = 0.3                # clipping range
TEMPERATURE = 0.95           # sampling temperature
BETA = 0.001                  # KL penalty (0 = disabled)
```

The main changes I made was to increase the number of generations. With only 2 generations, the average reward is too varied to improve the model. To fix this, I increased the generations to 6, then to 8, each addition increasing the final accuracy on the testset. I also increased the temperature to 0.95 to increase the creativity and randomness of responses, so that the average reward is more representative of all possible outputs. I also added a small KL penalty to promote regularization. The reward values slowly increased, although with high volatility. Some inputs improved the loss, some worsened the loss, but in general the effect was positive.

2. Compare the training dynamics of GRPO (this homework) with SFT/LoRA (HW3). Which converged faster, and which produced better results on your dataset?

The SFT approach had a difficult time converging for my previous use case (long-form responses), and it definitely converged slower. In general, this might be because of the difficulty of the task. However, I noticed that SFT had a harder time with consistent formatting, while GRPO almost immediately improved the formatting substantially. Also, the GRPO model did well after training on a specific question type, but was sensitive to the drift of the RL dataset over time, as it got worse during some iterations.

Problem 8: Post-Training Evaluation (20 points)

Problem 8.1 Load the Trained Model

Load the GRPO-trained LoRA adapters onto the base model for inference.

```
In [14]: from peft import PeftModel
from transformers import AutoProcessor, AutoModelForImageTextToText
import torch

MODEL_ID = "Qwen/Qwen3-VL-2B-Instruct"
OUTPUT_DIR = "grpo-output-gen8"

base_model = AutoModelForImageTextToText.from_pretrained(
```

```

MODEL_ID,
torch_dtype=torch.bfloat16,
device_map="auto",
)

ft_model = PeftModel.from_pretrained(base_model, OUTPUT_DIR)
ft_model.eval()

processor = AutoProcessor.from_pretrained(MODEL_ID)
print("GRPO-trained model loaded. Ready for inference.")

```

``torch_dtype` is deprecated! Use `dtype` instead!`

```

Loading weights:  0%|          | 0/625 [00:00<?, ?it/s]
GRPO-trained model loaded. Ready for inference.

```

Problem 8.2: Test on Held-Out Images (15 points)

Test the GRPO-trained model on your held-out test images. If you are using the default example dataset, the demo test samples below will work out of the box.

For each test image, the cell will:

1. Run inference with the same instruction suffix used during training
2. Extract the answer from after **Answer:**
3. Compare to ground truth

```

In [15]: import io, requests
from PIL import Image
from IPython.display import display

# =====
# ##### CHANGE ME #####
# =====

df_test = pd.read_json("/content/movie-dataset-creation/data_test.jsonl", lines=True)[:5]

# Replace with your own test samples, or use the defaults below.
# Each entry needs an image path (local file) or URL, a question, and ground truth.
TEST_SAMPLES = df_test.to_dict(orient='records')

MAX_NEW_TOKENS = 256

# =====
# ##### END CHANGE ME #####
# =====

INSTRUCTION_SUFFIX = (
    "\n\nFirst, think through your reasoning step by step. "
    "Then, provide your final answer as a single word on the last line after \"Answer:\"
    "For example:\nI can see a furry animal curled up on the couch...\nAnswer: cat"
)

def load_image(source: str) -> Image.Image:
    """Load image from a local path or URL."""
    if source.startswith("http://") or source.startswith("https://"):
        response = requests.get(source, stream=True)
        return Image.open(io.BytesIO(response.content)).convert("RGB")
    else:
        return Image.open("/content/movie-dataset-creation/" + source).convert("RGB")

```

```

results = []

for sample in TEST_SAMPLES:
    try:
        image = load_image(sample["image"])
    except Exception as e:
        print(f"Could not load image {sample['image']}: {e}")
        continue

    messages = [
        {"role": "user", "content": [
            {"type": "image", "image": image},
            {"type": "text", "text": sample["question"] + INSTRUCTION_SUFFIX},
        ]},
    ]

    text = processor.apply_chat_template(messages, tokenize=False, add_generation_prompt=True)
    inputs = processor(images=[image], text=[text], return_tensors="pt", padding=True).to(device)

    with torch.no_grad():
        output_ids = ft_model.generate(**inputs, max_new_tokens=MAX_NEW_TOKENS, temperature=TEMPERATURE)

    generated = processor.decode(output_ids[0][inputs["input_ids"].shape[1]:], skip_special_tokens=True)
    extracted = extract_answer(generated)

    results.append({
        "image": sample["image"],
        "question": sample["question"],
        "model_output": generated,
        "extracted_answer": extracted,
        "ground_truth": sample["answer"],
        "correct": (extracted is not None) and (extracted.lower().strip() == sample["answer"].lower().strip()),
        "has_format": extracted is not None,
    })

    display(image.resize((256, 256)))
    print(f"Question: {sample['question']}")
    print(f"Model output: {generated}")
    print(f"Extracted: {extracted}")
    print(f"Ground truth: {sample['answer']}")
    print(f"Correct: {results[-1]['correct']} | Format: {results[-1]['has_format']}")
    print("-" * 60)

n = len(results)
if n > 0:
    acc = sum(r["correct"] for r in results) / n
    fmt = sum(r["has_format"] for r in results) / n
    print(f"\nSummary: Accuracy={acc:.1%}, Format compliance={fmt:.1%} ({n} samples)")

```



Question: What is the primary genre of this movie?

Model output: I need to analyze the movie poster to determine its primary genre.

1. The title of the movie is "Demon Slayer: Infinity Castle". This is a well-known anime series.
2. The artwork features characters in intense combat, with one character holding a sword and another in a dynamic pose, suggesting action.
3. The visual style is highly detailed and dramatic, with vibrant colors and energy effects, which are typical of action anime.
4. The central theme revolves around a demon slayer fighting against powerful enemies, which is a core element of the genre.
5. The overall tone is one of intense struggle and high stakes, which is characteristic of the action genre.

Based on these observations, the primary genre of this movie is action.

Answer: action

Extracted: action

Ground truth: action

Correct: True | Format: True



Question: When was this movie released (decade)?

Model output: I need to determine the release decade of the movie "Starbright" based on the information provided.

1. The image is a movie poster for "Starbright".
2. The title is displayed at the bottom of the poster.
3. The poster shows a woman in a field under a starry sky, with a farmhouse in the distance.
4. The release year of a movie is typically associated with the decade it was released.
5. The movie "Starbright" was released in 2022.
6. The year 2022 falls within the 2020s decade.
7. The 2020s decade is the 2020s.

Answer: 2020s

Extracted: 2020s

Ground truth: 2020s

Correct: True | Format: True



Question: In what decade was this film released?

Model output: The movie "Predator: Badlands" is a 2020 film. This places its release in the 2020s, which is the 2020s decade.

Answer: 2020s

Extracted: 2020s

Ground truth: 2020s

Correct: True | Format: True



Question: Which decade is this movie from?

Model output: To determine the decade of the movie, let's analyze the information provided in the image.

1. The image is a promotional poster for a movie titled "Spider-Man: Into the Spider-Verse".
2. The release year of this movie is 2018.
3. The year 2018 falls within the 2010s decade.

Therefore, the movie is from the 2010s.

Answer: 2010s

Extracted: 2010s

Ground truth: 2010s

Correct: True | Format: True



Question: In what decade was this film released?

Model output: To determine the decade in which the film was released, let's analyze the movie's title and the characters depicted.

The movie title is "Deadpool & Wolverine". This title is a direct reference to the two main characters from the Marvel Comics universe. Deadpool is a character created by writer Chris Claremont and artist John Romita Jr., first appearing in 1991. Wolverine, a character created by writer Chris Claremont and artist John Byrne, first appeared in 1983.

The film "Deadpool & Wolverine" is a 2018 release. It is a film produced by Marvel Studios and directed by the Russo brothers, and it features the characters Deadpool and Wolverine in a crossover storyline.

The 2010s decade includes the years 2010 to 2019. Since the film was released in 2018, it falls within this decade.

Answer: 2010s

Extracted: 2010s

Ground truth: 2020s

Correct: False | Format: True

Summary: Accuracy=80.0%, Format compliance=100.0% (5 samples)

Problem 8.3: Reflection (5 points)

Compare the GRPO-trained model with the pre-trained base model (zero-shot) and your SFT model from HW3:

- Does the GRPO model use the `Answer:` format more consistently?

The GRPO model definitely formatted the answer more consistently than SFT or the base model. In the examples shown above, the formatting is perfect (though N=5). The base model formatted responses sometimes, but the SFT degraded formatting due to the open-endedness of the my previous dataset's answers. Because the supervision was over long sequences, formatting consistency became a much harder problem. The GRPO improved consistently almost immediately.

- Does it show step-by-step reasoning before the answer?

Yes, in almost all cases, after GRPO, there is successful step-by-step reasoning. There is a range of quality, from a bullet-point list of reasons, to more abstract chain-of-thought reasoning, but in general the reasoning is functional. In the SFT case, the reasoning is weak and inconsistent, and the training often lead to repeated cycles of phrases, not good, structured reasoning. The base model is able to do some basic reasoning, but is not as consistently strong as post-GRPO.

In []: